

Steven Flynn

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<https://stevenpatrickflynn.github.io/>

Research Interests

Hypoelliptic and Dispersive PDEs. Analysis on Lie groups. Microlocal Analysis. Geometric Inverse Problems. Sub-Riemannian geometry.

Employment

2024–present	Postdoctoral Fellow , University College London
2023–2024	Postdoctoral Fellow , University of Padova
2021–2023	Postdoctoral Fellow , University of Bath
2014–2020	Graduate Student Instructor & Graduate Teaching Assistant , University of California, Santa Cruz

Education

2014–2020	Ph.D. Mathematics , University of California, Santa Cruz (UCSC) <i>Dissertation title: “Unraveling Geodesic X-ray Transforms on the Heisenberg Group.”</i> <i>Co-advisors:</i> François Monard, Richard Montgomery
2014–2015	M.A. Mathematics , UCSC
2012–2014	B.A. Mathematics , UCSC

Research Papers

Journal Articles

- [1] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Quantization on filtered manifolds. arXiv:2412.17448, 2024. To appear in *Memoirs of the European Mathematical Society*
- [2] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Geometric invariance of the semi-classical calculus on nilpotent graded Lie groups. *Journal of Geometric Analysis*, 2023.
- [3] Steven Flynn. Injectivity of the Heisenberg X-ray transform. *Journal of Functional Analysis*, 280.5 (2021): 108886.

Refereed Proceedings and Volumes

- [4] Steven Flynn. Singular Value Decomposition of the X-ray Transform on the Reduced Heisenberg group, and a Two-Radius Theorem. Springer Volume, Trends in Mathematics: Ghent Analysis and PDE Center, 2023.
- [5] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Some Remarks on Semi-Classical Analysis on Two-Step Nilmanifolds. Springer INdAM Series, 2023.

Preprints

- [6] Davide Barilari and Steven Flynn. Refined Strichartz estimates for sub-Laplacians in Heisenberg and H-type groups. arXiv:2501.04415, 2025.
- [7] Steven Flynn. The sub-Riemannian X-ray Transform on H-type Groups: Fourier Slice Theorems and Injectivity Sets. arXiv:2312.00594, 2023.

Talks

Strichartz Estimates on H-type groups

May 2025 UC Santa Cruz
 Jan 2025 Lund University
 Dec 2024 University College London

A Microlocal Calculus on Filtered Manifolds

Dec 2025 HADES Seminar, UC Berkeley
 Dec 2025 Geometric Structures Seminar, SISSA
 Jan 2025 University College London
 May 2024 Dispersion and Geometry in Padova

The sub-Riemannian X-ray Transform on H-type groups

Nov 2023 Junior Methusalem Seminar, Ghent University
 Oct 2023 University College London
 Nov 2022 Geometry and Analysis Seminar, University of Bristol

Remarks on the Semi-classical analysis of Step 2 Nilpotent groups

May 2023 Operator Algebras in the South of the UK, University of Southampton
 Mar 2023 CAGE mini-seminar, Sorbonne Université

Geometric Invariance of the Semi-Classical Calculus on Graded Lie groups

Nov 2022 Conference on Noncommutative Analysis and PDEs, Queen Mary University

The Heisenberg X-ray transform

May 2022 Analysis and Differential Geometry International Seminar, University of Aveiro
 Mar 2022 AGeNT Seminar, University of Bath
 Nov 2021 Problèmes Spectraux en Physique Mathématique, Institut Henri Poincaré
 Sep 2021 Bath Analysis Seminar, University of Bath
 Apr 2021 Sub-Riemannian Seminars (online)
 May 2020 UCSC Geometry and Analysis Seminar

Integral Geometry on Contact Manifolds

Jan 2020 Joint Math Meeting, Denver, Colorado
 Nov 2019 Mathematical Sciences Research Institute, Berkeley, CA
 Sep 2019 MSRI, Berkeley, CA

Service

Reviewer, *Journal of Functional Analysis* 2022–2023
 Organizer, Bath Analysis Seminar 2022
 Organizer, University of Bath Postdoc Away Day 2022
 Organizer, Graduate Student Seminar, MSRI 2019
 Organizer, Microlocal Analysis Seminar, UCSC 2018
 Organizer, Graduate Differential Geometry Seminar, UCSC 2018

Professional Activities

Participant, MPS Workshop on LEAN 2025
 Program Associate, MSRI Microlocal Analysis Program 2019
 Mentor, UCSC Directed Reading Program 2018

Teaching

Taught Course Centre (Bath, Bristol, Imperial, Oxford, Swansea, Warwick)

Autumn 2026 Analysis of the Sub-Laplacian (Term 1)

University College London

Autumn 2025 MATH0071: Spectral Theory

Supervision

2026–27	MSci project: degenerate diffusions and hypoellipticity, UCL
Aug–Oct 2026	Summer research project: equidistribution and ergodic theorems, UCL
2018	Mentor, Directed Reading Program, UCSC

University of California, Santa Cruz

Summer 2019	Math 105A: Real Analysis
Spring 2019	Math 105B: Real Analysis II
Summer 2018	Math 105A: Real Analysis
Summer 2017	Math 3: Precalculus

Graduate Teaching Assistant

Six years of continuous teaching experience (20 hours/week). Duties included delivering lectures and review sessions, grading homework and examinations, and holding office hours.

2020 Spring	Math 105A: Real Analysis
2020 Winter	Math 105B: Real Analysis II
2019 Fall	Math 23B: Vector Calculus II
2019 Spring	Math 105B: Real Analysis II
2019 Winter	Math 105A: Real Analysis
2018 Fall	Math 152: Programming for Math
2018 Spring	Math 19B: Calculus for Science, Engineering and Mathematics II
2018 Winter	Math 100: Introduction to Proofs and Problem Solving
2017 Fall	Math 19A: Calculus for Science, Engineering and Mathematics
2017 Spring	Physics 6/6L: Introductory Physics Laboratory
2017 Winter	Math 23A: Vector Calculus
2016 Fall	Math 2: College Algebra (Collaborative Learning version)
2016 Spring	Physics 6/6L: Introductory Physics Laboratory
2016 Winter	Math 23A: Vector Calculus
2015 Fall	Math 3: Precalculus
2015 Spring	Math 11A: Calculus with Applications
2015 Winter	Math 21: Linear Algebra
2014 Fall	Math 2: College Algebra